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Job Scheduling Algorithm for a Hybrid MTO–MTS Production Process

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Abstract: Make-to-Order (MTO) manufacturing (especially in the case of seasonal production) creates problems associated with the low use of production capacity in periods of low demand. In order to avoid unnecessary costs, companies often choose to manufacture selected products to stock. However, in doing so they face important decision problems related to the selection of components that will be made to stock (MTS) and designing a production schedule that will take into account both current production orders as well as MTS operations. In this study, we present the assumptions of a method and a main algorithm for production job scheduling in hybrid MTO-MTS production systems. This paper presents both the individual stages of the procedure as well as an example illustrating the operation of the algorithm.

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Keywords: hybrid production, MTO-MTS, Make-to-Stock, Make-to-Order, scheduling, production planning, production strategy

1. INTRODUCTION

The growing, global competition means, that apart from low-cost manufacturing of products of appropriate quality (Relich and Świć, 2020; Saha at al. 2021), also the ability to quickly (and most of all timely) complete production orders becomes crucial for the development options of enterprises (Wikarek and Sitek 2021). One of the enterprises most important task is active developing their processes (Krajcovic and Plinta, 2012; Wiecek et al. 2020) by actively using optimisation tools (Krajcovic et al. 2021). Manufacturing companies always seek to maximize the use of their human potential and the production capacity of their machines and devices (Kłosowski et al. 2015; Grznar et al. 2020), trying, at the same time, to minimize production costs (Gola, 2021; Danilczuk and Gola 2020, Relich 2016) and reach sustainability challenges (Nielsen et al. 2020). Given these goals, the key decision they have to make is to select an appropriate production strategy (Grzelak et al. 2022, Kotowska et al. 2018).

The literature most often mentions two classic production strategies, known as Make-to-Order (MTO) and Make-to-Stock (MTS) manufacturing (Kuthambalayand and Bera, 2020a) – Table 1. For some time now, both in the literature and in industrial practice, there has been talk of hybrid production strategies, among others, strategies combining MTO with MTS (Xiong et al. 2020, Fiems et al. 2020), MTO with Make-to Availability (MTA) (Ciechańska and Szwed 2020), and Mass Customization with Engineer-to-Order production (MC-ETO) (Ciesla and Mleczko, 2021). However, these approaches have not yet been subject to extensive academic discussion, and the concept of a hybrid production strategy or a hybrid production model does not have an unambiguous definition and appears in the literature in various contexts (Peeters & van Ooijen, 2020, Antosz et al. 2021).

Table 1. Comparison of MTO and MTS production strategies

Feature	MTS	MTO
Goal	to maintain a certain amount of finished goods inventory	to fulfil a customer order within a specified time
Production planning approach	production is planned on the basis of demand forecasts and inventory turnover	short-term planning is based on orders received, long-term planning is based on sales plans
Start of production	production begins when the inventory on hand drops below the minimum level	production begins when a customer order has been received
Product	standard, repeatable	bespoke-designed, bespoke-assembled, or bespoke-finished
Production volume	mostly mass, high-volume or medium-volume production	mostly small-volume or one-off production
Industries and applications	FMCG, furniture, electronics and household appliances	machinery, construction, IT systems, exclusive products
Variants of manufacturing models	make-to-availability (MTA)	assemble-to-order (ATO), finish-to-order (FTO), engineer-to-order (ETO)

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When analyzing the conditions in which companies (especially those in the SME sector) have to operate nowadays, one notices that it is more and more difficult to run production using only one of the classic manufacturing models and that the total production cost can only be reduced by using mixed solutions combining the features of MTO and MTS production (Kuthambalayand and Bera, 2020b). A classic example are businesses that normally use the MTO production strategy, but in periods when order volumes low, are forced to temporarily make products to stock in order to fully use their production capacity (Yousefnejad and Esmaeili, 2020; Jasiulewicz-Kaczmarek 20218). The key decision problems they have to face regard the selection of the components to be produced and proper scheduling of jobs allowing to maximize the use of machinery (Bocewicz et al, 2008; Klos and Patalas-Maliszewska, 2017; Gola et al, 2021).

In this article, we present a general algorithm for scheduling MTO-MTS production jobs. In particular, we provide a definition and describe the features of hybrid MTO-MTS production (Section 2), present the main assumptions of the algorithm developed in this study (Section 3), and give an example illustrating how this solution works (Section 4).

2. CHARACTERISTICS OF HYBRID MTO-MTS PRODUCTION

Hybrid MTO-MTS production can be modelled as production in which a company produces part of its product range using the MTO approach and part using the MTS approach. There is no rigid definition of what features production under each of these strategies should have and what quantities of goods should be produced using each approach for the production process to be called a hybrid process. For instance, (Zhang et al., 2013) illustrate the concept of hybrid production, describing a case of a footwear manufacturing company in which customer orders for standard products are fulfilled on an MTS basis, while customized production orders are met in an MTO process. A similar definition can be found in (Soman et al., 2004), which gives the example of the food manufacturing industry, in which, depending on the requirements regarding packaging, products are either delivered directly from a warehouse or produced to order. In (Jiang & Wu, 2021), in which the problem is defined similarly, the authors focus on job flow through the manufacturing environment. A slightly different view of the hybrid production model is presented by Wu et al. (Wu et al., 2008). They discuss an example in which a foundry manufactures semi-finished products to stock in periods when the number of orders declines, because this is more economically profitable than reducing and then again restoring production capacity – an assumption that we also adopt in this present paper.

An interesting overview of assumptions for hybrid production models can also be found in a study by Peeters and Van Ooijen (Peeters & Van Ooijen, 2020). In this present article, hybrid MTO-MTS production is understood similarly to (Adan & Van Der Wal, 1998; Morikawa et al., 2014), i.e. as a process in which some semi-finished products can be manufactured to stock and used later as components of machines made to order. Components manufactured in advance can also be used as spare parts in the event of failure of already produced machines.

A key problem one encounters when trying to conduct a comprehensive review of the literature on hybrid production is a lack of established terminology. The most commonly used keywords are "make-to-stock", "make-to-order" and "production control", but entering them in the search on their own makes it difficult to find texts devoted to mixed MTO-MTS production. Table 2 lists the search terms used in our search along with the number of publications containing those terms available in the Scopus database (as of 31/07/2021). Figure 1 shows the number of publications devoted to the hybrid production model referenced in (Peeters and van Ooijen, 2020).

Table 2. Number of Scopus publications for the search terms used in this study

Search term	Number of publications
“Make to order” and “Make to stock”	1136
“Make to order” and “Make to stock” and “production control”	131
“Make to order” and “Make to stock” and “production planning”	58
“Make to order” and “Make to stock” and “scheduling”	40
“Make to order” and “Make to stock” and “production strategy”	12
“hybrid MTO-MTS”	4

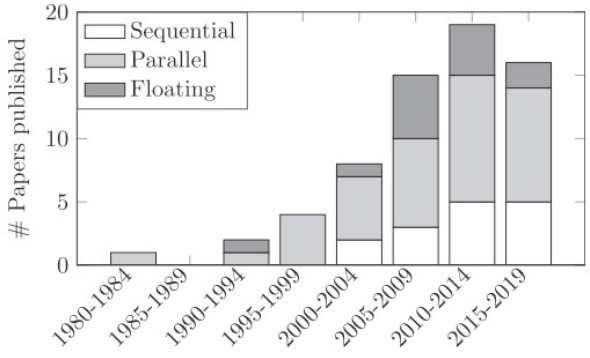


Figure 1. Number of publications on hybrid production

The review of the literature showed that, unfortunately, there were few studies related to the problem of job scheduling for hybrid production. Worth noting and recommending is a paper by (Chang et al., 2003), in which the authors developed a production planning algorithm that took into account due dates for MTO products, at the same time allowing to fill production capacity with MTS components (details). The authors compared the results obtained using their algorithm with known priority rules and heuristics. In this present paper, we discuss a similar case of production planning and a similar methodology for validating the quality of the proposed algorithm. Our proposal, however, differs from Chang et al.'s in that we adopt different assumptions regarding the hybrid

model (we define a specific case of hybrid production in a different way) and use distinct decision criteria for the proposed algorithm.

3. JOB SCHEDULING IN HYBRID PRODUCTION

3.1 Definition of the research problem

When delineating the scope of the present study, we defined hybrid production as a case in which a company produces parts using the MTO strategy and, additionally, fills the production plan with MTS semi-finished products and parts. For this model, we developed a job scheduling algorithm for production scheduling (Fig. 2).

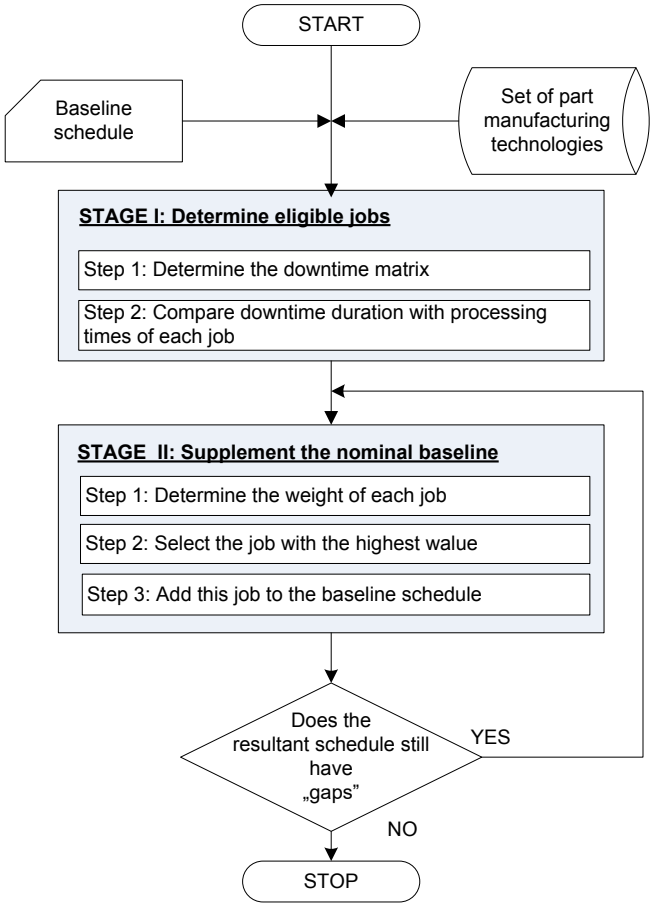


Figure 2. A general job scheduling algorithm for hybrid MTO-MTS production

When developing this solution, we assumed the following:

1. The initial production planning model is an MTO production model. Production is planned assuming that products must be manufactured within a predefined time horizon (planning period). As a result of planning, a *baseline schedule* is created.
2. Once production to order has been planned, the company still has free processing capacity that it wants to use to make semi-finished products to stock.
3. The company has a specific set of parts that it can produce under the MTS approach. It must decide which of the jobs can be added to the baseline schedule so as to minimize the cost of storage and the risk of not selling the products.
4. In designing a schedule for MTS parts (hereinafter referred to as *supplemented schedule*), the due dates for MTO production jobs must not be changed.
5. Jobs are scheduled as part of a *Job Shop Scheduling Problem* (JSSP), jobs are indivisible, and job times are deterministic.

3.2 Input data

The input data to the scheduling algorithm developed in this study include the *baseline schedule* with a specific time horizon and a set of jobs that the company can produce using the MTS strategy, hereinafter referred to as the *MTS job set* (they are parts that are universal for all the finished products offered by the company and typical service and fast-wearing parts for the machines manufactured by the company). The baseline schedule does not fully use the company's production capacity (Line Efficiency coefficient *LE* <100%) and so a job from the MTS job set can be added to it.

3.3 First stage of the algorithm – determination of eligible jobs

The first stage of the algorithm is to determine which of the jobs from the MTS job set are eligible to be processed (i.e. which jobs can be added to the baseline schedule). To find these jobs, the algorithm performs a series of operations. The first one is to determine the *downtime matrix.* It is a matrix that defines all downtime times (downtime start time, downtime duration, and downtime end time) for each machine M. Machine downtime represents unused production potential that can be tapped by supplementing the schedule with production jobs from the MTS job set. The second stage is to compare downtime times on machines M with the processing times on these machines required to perform the individual jobs from the MTS job set (the *processing time matrix*). For example, machine M1 has a downtime duration of 2 units of time and machine M2 has a downtime duration of 4 units of time. J1 requires 2 units of processing time on M1 and 3 units of processing time on M2 (see Section 4).

For those jobs from the MTS job set whose processing times on the individual machines are equal or shorter than the downtime on these machines, the algorithm checks the technological route (the *machine order matrix*). In the example given above, it may turn out that the downtimes on machines M1 and M2 overlap; in a situation like that, job J1 must first be processed on M2 before it can be moved to M1 and so it cannot be added to the schedule due to J1 sequencing constraints and downtimes on machines M1 and M2.

After these steps have been completed, the jobs from the MTS job set which can be placed in the baseline schedule (eligible MTS jobs) are determined as the product of the processing time matrix and the machine order matrix.

3.4.Stage two of the algorithm – supplementation of the baseline schedule

Once the eligible MTS jobs have been identified, the algorithm has to decide which of these jobs to place in the baseline

schedule. To this end, a weighting parameter is determined for the jobs from this set, taking into account the availability and costs of materials, the costs of frozen capital and the chance that the given semi-finished product will be used in the shortest possible time. Due to length restrictions, the weight calculation algorithm is not described in detail in this paper. Jobs with assigned weights are placed in the *weights matrix.*

After the weighting procedure has been completed, the algorithm selects, from the set of eligible MTS jobs, a job with the highest weight. Next, this job is added to the baseline schedule in the first available slot, in accordance with the available machine capacity (downtime) and the job's machine order.

As a result of the algorithm having completed the loop, a supplemented schedule is created, which is a baseline schedule to which one eligible MTS job has been added. The supplemented schedule utilizes the company's production capacity to a greater extent than the baseline schedule (LE of the supplemented schedule> LE of the baseline schedule).

3.5 Further iterations of the algorithm

The first iteration of the algorithm only allows one MTS job to be added to the baseline schedule. One may find, however, that the downtime times in the baseline schedule make it possible to add more than one MTS job to that schedule. That is why the hybrid production planning algorithm proposed in this paper has been designed as an iterative algorithm. After the first loop has been completed, the original baseline schedule is overwritten by the supplemented schedule. The entire algorithm loop is completed again for the new baseline schedule (which already contains one MTS job). The algorithm iterates until none of the jobs in the MTS job set can be placed in the schedule (i.e. until the set of eligible MTS jobs becomes an empty set).

Each successive iteration of the algorithm increases the utilization of the company's production capacity (the LE of each subsequent schedule is greater than that of the previous one). The number of iterations also affects the weight calculating algorithm (for example, after 3 iterations, it may turn out that the company does not have the materials to make a given semi-finished product, even though it has enough downtime).

4. PROOF OF CONCEPT

The input data for the algorithm (described in Section 3) include the baseline schedule (Fig. 3) and the MTS job set. The MTS job set is described in the form of the machine order matrix (Table 3) and the processing time matrix.

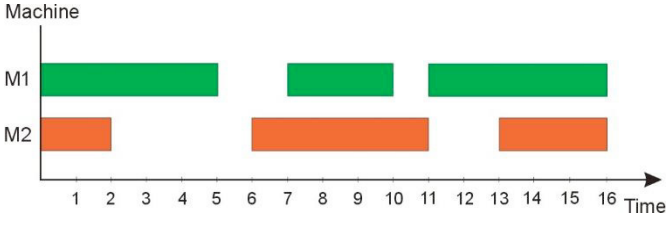


Figure 3. Baseline schedule.

Table 3. Machine order of the jobs in the MTS job set

	M1	M2
J1	2	1
J2	1	0
J3	0	1
J4	1	2
J5	2	1

Table 4. Processing time of the jobs in the MTS job set

	M1	M2
J1	2	3
J2	4	0
J3	0	2
J4	1	2
J5	2	1

Then the algorithm goes to the stage of determining the eligible jobs (described in Section 3.3). Step one – the algorithm determines a downtime matrix for each machine (Table 5).

Table 5. Downtime matrix

	Downtime start time	Downtime duration	Downtime end time
M1	5	2	7
M1	10	1	11
M2	2	4	6
M2	11	2	13

Step two – the algorithm defines the processing time matrix (Table 6) and the machine order matrix (Table 7).

Table 6. The processing time matrix specifies whether job processing times are less than or equal to downtime duration

J1	1
J2	0
J3	1
J4	1
J5	1

Table 7. The machine order matrix specifies whether the job execution order is consistent with the order of downtime on the individual machines and the duration of these downtimes

J1	0
J2	0
J3	1
J4	1
J5	1

Once the processing time matrix and the machine order matrix are known, the algorithm calculates the eligible jobs matrix as a product of the previously calculated matrices (Table 8).

Table 8. Eligible jobs matrix

J1	0
J2	0
J3	1
J4	1
J5	1

Next, the algorithm goes to the stage of supplementing the baseline schedule (described in Section 3.4). The first step is to calculate the weights matrix according to the weight calculation algorithm (Table 9).

Table 9. Weights matrix

J3	8
J4	3
J5	10

Job J5 is eligible and has the highest weight. It is added to the baseline schedule, as a consequence of which, a supplemented schedule is created (Fig. 4).

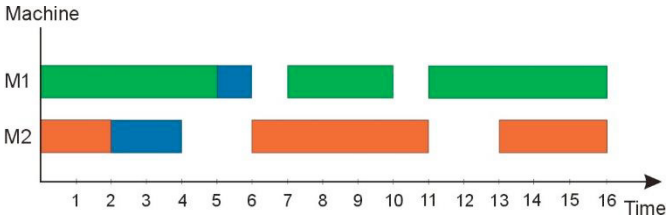


Figure 4. Supplemented schedule after the first iteration.

The supplemented schedule is overwritten as a baseline schedule, and the algorithm performs the next iteration (as described in Section 3.5) – Fig. 5.

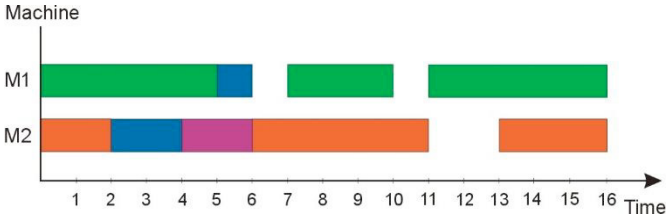


Figure 5. Supplemented schedule after the second iteration.

After 3 iterations, a supplemented schedule is created which cannot be supplemented with further jobs from the MTS job set (Fig. 4). In the case under study, the LE of the supplemented schedule was 96.875%, and was higher than the LE of the baseline schedule, which was 71.875%.

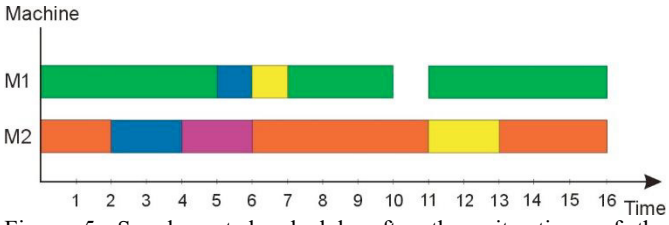


Figure 5. Supplemented schedule after three iterations of the algorithm.

5 CONCLUSIONS

Hybrid MTO-MTS production generates decision problems regarding both the selection of components to be produced to stock and appropriate scheduling of production by supplementing the baseline schedule of MTO jobs with MTS jobs. In this paper, we proposed a shop job scheduling approach that provides both for selecting jobs to be produced using the MTS strategy and scheduling those jobs. Components for MTS production were selected under the assumption of minimizing the total cost of storage and maximizing the probability that the manufactured products will be sold. In consequence the method gives possibility to increase the level of machines capacity utilization and therefore reduce "empty costs" of production process. The proposed algorithm allows to fill in the initial MTO algorithm without necessity of optimizing all production tasks what makes it useful for small and medium companies.

In the future, we are going to do research regarding detailed parametrization of the algorithm for selecting components for MTS production and to compare the proposed job scheduling approach with classic priority rules (LPT, SPT, etc.). Moreover we will analyze the cost of production and storage

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